

## Exercise Sheet 4

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**Due date: 15:30, May 26th.**

**Late submissions will receive no credit.**

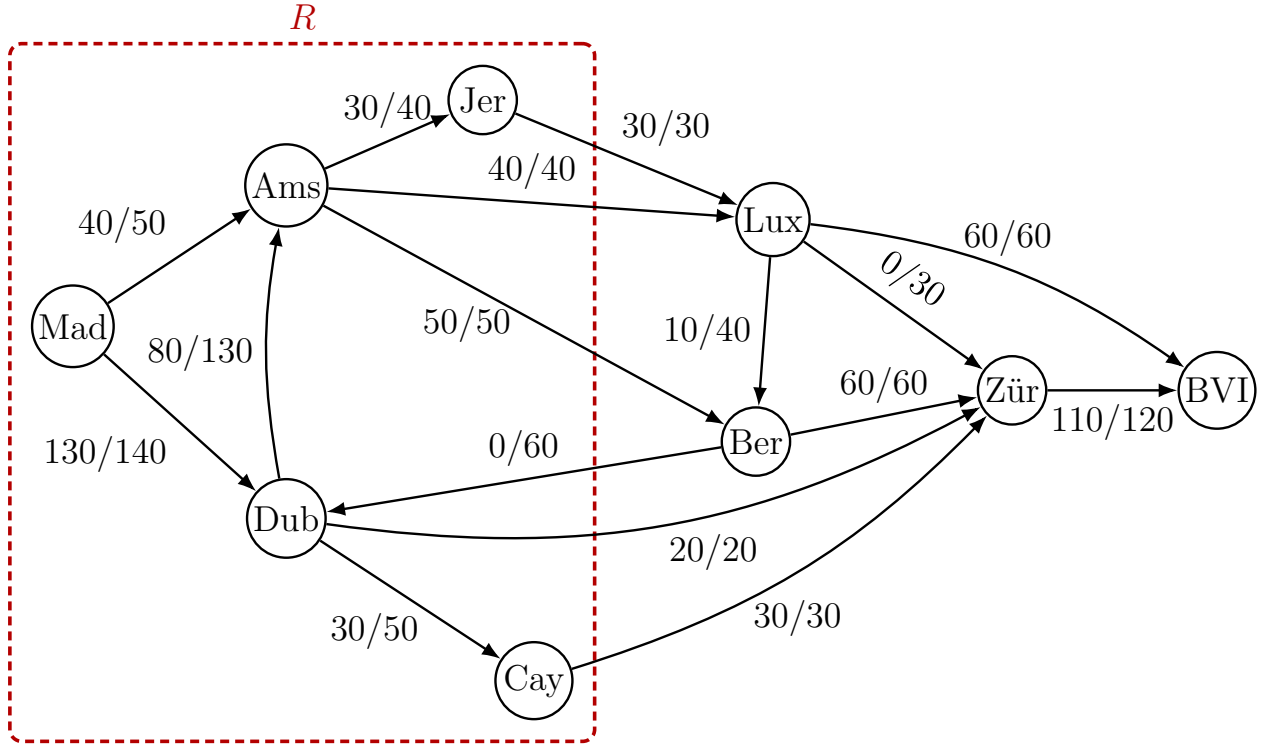
You should try to solve all of the exercises below, and then submit two solutions to be graded — each problem is worth 10 points. Make sure to clearly justify your answers, defining everything precisely and stating any results you are using. You should then submit your solutions on NTU COOL, either as a typed PDF or a clear picture or scan of your handwritten answers.

**Exercise 1** Table 1 provides the capacities along the edges of a network. Find a flow of maximum possible value from Madrid (‘Mad’) to the British Virgin Islands (‘BVI’), and give a short proof that one cannot do better.

| Start | → | End | Capacity (\$, mil) | Start | → | End | Capacity (\$, mil) |
|-------|---|-----|--------------------|-------|---|-----|--------------------|
| Mad   | → | Ams | 50                 | Mad   | → | Dub | 140                |
| Ams   | → | Jer | 40                 | Ams   | → | Lux | 40                 |
| Ams   | → | Ber | 50                 | Jer   | → | Lux | 30                 |
| Dub   | → | Ams | 130                | Dub   | → | Zür | 20                 |
| Dub   | → | Cay | 50                 | Ber   | → | Dub | 60                 |
| Ber   | → | Zür | 60                 | Lux   | → | Ber | 40                 |
| Lux   | → | Zür | 30                 | Lux   | → | BVI | 60                 |
| Cay   | → | Zür | 30                 | Zür   | → | BVI | 120                |

Table 1: The capacities along the edges of a network.

**Solution:** A maximum flow has value 170. One such flow is shown below; each edge label has the form  $f/c$ , where  $f$  is the flow on that edge and  $c$  is its capacity.



The corresponding edge flows are:

| edge                  | $f$ | edge                  | $f$ |
|-----------------------|-----|-----------------------|-----|
| Mad $\rightarrow$ Ams | 40  | Mad $\rightarrow$ Dub | 130 |
| Ams $\rightarrow$ Jer | 30  | Ams $\rightarrow$ Lux | 40  |
| Ams $\rightarrow$ Ber | 50  | Jer $\rightarrow$ Lux | 30  |
| Dub $\rightarrow$ Ams | 80  | Dub $\rightarrow$ Zür | 20  |
| Dub $\rightarrow$ Cay | 30  | Ber $\rightarrow$ Dub | 0   |
| Ber $\rightarrow$ Zür | 60  | Lux $\rightarrow$ Ber | 10  |
| Lux $\rightarrow$ Zür | 0   | Lux $\rightarrow$ BVI | 60  |
| Cay $\rightarrow$ Zür | 30  | Zür $\rightarrow$ BVI | 110 |

All listed flows are at most their capacities. Flow is also conserved at each intermediate vertex: for example, Ams receives  $40 + 80 = 120$  and sends  $30 + 40 + 50 = 120$ , while Zür receives  $20 + 60 + 0 + 30 = 110$  and sends 110. The value of the flow is therefore

$$40 + 130 = 170,$$

the total flow leaving Mad, equivalently  $60 + 110 = 170$ , the total flow entering BVI.

It remains to show that no larger flow is possible. Consider the cut

$$R = \{\text{Mad, Ams, Dub, Jer, Cay}\}, \quad V \setminus R = \{\text{Lux, Ber, Zür, BVI}\}.$$

The edges directed from  $R$  to  $V \setminus R$  are

$$\text{Ams} \rightarrow \text{Lux}, \quad \text{Ams} \rightarrow \text{Ber}, \quad \text{Jer} \rightarrow \text{Lux}, \quad \text{Dub} \rightarrow \text{Zür}, \quad \text{Cay} \rightarrow \text{Zür}.$$

Their total capacity is

$$40 + 50 + 30 + 20 + 30 = 170.$$

Every feasible flow is bounded above by the capacity of every source-sink cut, so no flow can have value more than 170. Since the displayed feasible flow has value 170, the maximum possible flow is

$$\boxed{170 \text{ million dollars}}.$$

## Exercise 2

- (a) If  $G$  is  $k$ -connected, and  $G'$  is obtained by adding a new vertex  $v$  of degree at least  $k$ , show that  $G'$  is also  $k$ -connected.
- (b) Prove that  $G$  is  $k$ -connected if and only if it has at least  $k + 1$  vertices and, for any two sets  $A, B \subset V(G)$  of size  $k$ , there are  $k$  pairwise-disjoint paths<sup>1</sup> that start in  $A$  and end in  $B$ .

## Solution:

- (a) We want to show that for any subset  $S \subseteq V(G')$  with  $|S| < k$ , the graph  $G' - S$  is connected. We consider two cases based on whether the new vertex  $v$  is removed:

**Case 1:**  $v \in S$ .

In this case, removing  $S$  from  $G'$  is equivalent to removing  $S \setminus \{v\}$  from  $G$ . That is,

$$G' - S = G - (S \setminus \{v\})$$

Since  $|S| < k$ , it follows that  $|S \setminus \{v\}| < k - 1 < k$ . Because  $G$  is  $k$ -connected, removing fewer than  $k$  vertices from  $G$  leaves it connected. Thus,  $G - (S \setminus \{v\})$  is connected, which implies  $G' - S$  is connected.

**Case 2:**  $v \notin S$ .

Since  $v \notin S$ , we can view  $G' - S$  as the graph  $G - S$  with the vertex  $v$  (and its remaining incident edges) added back. Because  $G$  is  $k$ -connected and  $|S| < k$ , the graph  $G - S$  is connected. Furthermore, we are given that  $\deg_{G'}(v) \geq k$ . Since  $|S| \leq k - 1$ , the vertex  $v$  loses at most  $k - 1$  neighbors when  $S$  is removed. Therefore, in  $G' - S$ , the vertex  $v$  retains at least:

$$\deg_{G' - S}(v) \geq k - |S| \geq 1$$

neighbor. Since  $G - S$  is connected and  $v$  has at least one neighbor in  $G - S$ , the graph  $G' - S$  remains connected.

Since  $G' - S$  is connected in both cases,  $G'$  is  $k$ -connected.

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<sup>1</sup>Note that a path could consist of just a single vertex.

- (b) ( $\Rightarrow$ ) Assume  $G$  is  $k$ -connected and  $|V(G)| \geq k + 1$ . Let  $A, B \subseteq V(G)$  be any two subsets of size  $k$ .

We construct an auxiliary graph  $G^*$  by adding two new vertices  $s$  and  $t$  to  $G$ , such that  $s$  is connected to all vertices in  $A$ , and  $t$  is connected to all vertices in  $B$ . Since  $\deg(s) = k$  and  $\deg(t) = k$ , applying the result from part (a) twice guarantees that  $G^*$  remains  $k$ -connected.

By the local-vertex version of Menger's Theorem, since  $s$  and  $t$  are non-adjacent vertices in a  $k$ -connected graph, there exist  $k$  internally vertex-disjoint paths between  $s$  and  $t$  in  $G^*$ . Each of these  $k$  paths must be of the form  $s - a_i - \dots - b_i - t$ , where  $a_i \in A$  and  $b_i \in B$ . By deleting the endpoints  $s$  and  $t$  from each path, we obtain  $k$  paths in  $G$  that start in  $A$  and end in  $B$ . Since the original paths were internally disjoint, these  $k$  resulting paths in  $G$  must be pairwise vertex-disjoint.

( $\Leftarrow$ ) Assume  $|V(G)| \geq k + 1$  and for every  $A, B \subseteq V(G)$  with  $|A| = |B| = k$ , there exist  $k$  pairwise-disjoint paths starting in  $A$  and ending in  $B$ . We show by contradiction that  $G$  is  $k$ -connected.

Suppose  $G$  is not  $k$ -connected. Then there exists a separating set  $S \subset V(G)$  with  $|S| < k$  such that  $G - S$  is disconnected. We can expand  $S$  to a set  $S'$  of size exactly  $k - 1$  by arbitrarily adding  $k - 1 - |S|$  vertices from the remaining components. Since  $|V(G)| \geq k + 1$ , removing  $S'$  leaves at least two vertices, and  $G - S'$  remains disconnected.

Let  $C_1$  and  $C_2$  be two distinct connected components of  $G - S'$ . Choose an arbitrary vertex  $x \in C_1$  and  $y \in C_2$ . Now, construct two sets  $A$  and  $B$  of size  $k$  as follows:

$$A = \{x\} \cup S' \quad \text{and} \quad B = \{y\} \cup S'$$

Note that  $|A| = |B| = k$  because  $x, y \notin S'$ .

By our initial assumption, there must exist  $k$  pairwise-disjoint paths from  $A$  to  $B$ . Since these  $k$  paths are pairwise vertex-disjoint, each of the  $k - 1$  vertices in  $S'$  can belong to at most one path. Therefore, at most  $k - 1$  of these paths can contain any vertex from  $S'$ .

Since there are  $k$  paths in total, at least  $k - (k - 1) = 1$  path must be completely free of any vertices in  $S'$ . This path must begin at the only remaining vertex of  $A$  (which is  $x$ ) and end at the only remaining vertex of  $B$  (which is  $y$ ).

Consequently, this path lies entirely within  $G - S'$ , establishing a connection between  $x \in C_1$  and  $y \in C_2$ . This directly contradicts the fact that  $C_1$  and  $C_2$  are disconnected components in  $G - S'$ .

Thus,  $G$  must be  $k$ -connected.

**Exercise 3** Consider the network in the figure below. The source and the sink are marked with  $S$  and  $T$ , and the capacities of all but one edge are indicated. The remaining edge has capacity  $\frac{1}{2}(\sqrt{5} - 1)$ .

- (a) Find (with proof) the value of the maximum flow in the network.
- (b) Describe a choice of augmenting paths in the Ford-Fulkerson algorithm for which the algorithm never finishes and the flow value converges to  $2 + \sqrt{5}$ .

**Exercise 4** The goal of this exercise is to prove the local, vertex version of Menger's Theorem; that is, given a graph  $G$  and non-adjacent vertices  $x, y \in V(G)$ ,  $\kappa(x, y) = \lambda(x, y)$ . We construct a directed graph  $\vec{D}$  with vertices  $V(\vec{D}) = \{v^+, v^- : v \in V(G)\}$  and edges  $\vec{E}(\vec{D}) = \{(u^+, v^-), (v^+, u^-) : uv \in E(G)\} \cup \{(v^-, v^+) : v \in V(G)\}$ . We then assign edge capacities by setting

$$c(\vec{e}) = \begin{cases} 1 & \text{if } \vec{e} = (v^-, v^+), v \in V(G) \\ \infty & \text{otherwise} \end{cases},$$

and consider flows in the network  $(\vec{D}, x^+, y^-, c)$ .

- (a) Prove the following claim: the minimum capacity of an  $x^+, y^-$  cut in  $\vec{D}$  is equal to the minimum size of an  $x, y$ -separating set in  $G$ .
- (b) Deduce the local vertex version of Menger's Theorem.